



CS-467- Software Project Management

Lecture No. 2

Software Project Evaluation Techniques

List of Topic to be Discuss in this lecture

- Effective Project Manager Competencies
- Code of Conducts and Ethical Aspects for Project Manager
- Four Project Dimensions
- How Project Manager Deal with Politics in Organization
- Project, Program And Portfolio Management
- Project Selection Methods

Effective Project Manager Competencies

1. **Knowledge:** What project manager **knows** about project?
2. **Performance:** What project manager is **able to do**, or accomplished with his PM Knowledge?
3. **Personal skills:** How the project manager **behaves** when performing the project or related activity?

Code of Conducts and Ethical Aspects for Project Manager

The code of conduct of the PMI has been written by practitioners and is organized in four areas:

- ❑ **Responsibility:** the duty of taking ownership of decisions made or failed to make and their consequences
- ❑ **Respect:** the duty of treating with respect the resources assigned to us, such as people, money, reputation, environment, and so on
- ❑ **Fairness:** the duty of taking decisions impartially and objectively.
- ❑ **Honesty:** the duty of acting in a truthful manner.

Four Project Dimensions

Four P's have a substantial influence on software project management—people, product, process, and project.

- ❑ **People** must be organized into effective teams, motivated to do high-quality software work, and coordinated to achieve effective communication.
- ❑ The **product** requirements must be communicated from customer to developer, partitioned (decomposed) into their constituent parts, and positioned for work by the software team.
- ❑ The **process** must be adapted to the people and the problem. A common process framework is selected, an appropriate software engineering paradigm is applied, and a set of work tasks is chosen to get the job done.
- ❑ The **project** must be organized in a manner that enables the software team to succeed.

How Project Manager Deal with Politics in Organization

- ❑ **Don't feed the fire.** When stakeholders try to get you on their side of an issue by complaining, gossiping, or moaning about a problem, a project solution, or another team member, don't agree or contribute.
- ❑ **If you can't say something nice, don't say anything at all.** Don't say anything negative about a stakeholder that you wouldn't say in his or her presence. Do share constructive criticism openly, and certainly do share compliments and kudos loudly and proudly.
- ❑ **Do stand up.** When you know you're right, don't succumb to what you know is wrong. Tact, honesty, and logic can go a long way here, but don't roll over just because someone higher in the organization has a differing opinion. It's okay to disagree, but only stand up when you know you have the ammo to back up your position. Also be sure to stand up and admit when you're wrong. This goes a long way to increasing your credibility and integrity.

How Project Manager Deal with Politics in Organization

- ❑ **Do protect your reputation.** When you get down to it, your reputation is important. You have to deliver on your promises, keep out of the gossip, treat everyone fairly, and focus on project deliverables.
- ❑ **Don't micromanage.** As a software project manager, chances are you've come up through the ranks. You've worked with different software languages, developed countless applications, and can out-code your project team members. But not anymore. Your job is to lead, manage, and provide guidance. You have to let your project team do the work.
- ❑ **Do play fair.** Some team members, stakeholders, and customers will drive you nuts, but you must treat everyone the same when it comes to discipline, time off, and kudos. If you're fair to your stakeholders they'll respect you — and respect is what you need the most to abstain from office politics.

Project, Program and Portfolio Management

Project

As we discuss in lecture #1 a project is a temporary endeavor undertaken to create a unique product, service, or result. They have a specific start and end date. A project is a small unit of large program or portfolio and is managed according to a common objective.

Program

Programs are grouped within a portfolio and are comprised of subprograms, projects, or other work that are managed in a coordinated fashion in support of the portfolio.

For example, an anonymous automobile company manufactures different types of cars. The design and manufacturing of one particular type of car would be called a program and of the parts like bonnet, panel, or engine would be a project. The integration of all the parts (projects) would make a complete automobile (program).

Project, Program and Portfolio Management

Program

A program is a group of projects that are similar or related to one another, and which are often managed and coordinated as a group instead of independently.

For example, an anonymous automobile company manufactures different types of cars. The design and manufacturing of one particular type of car would be called a program and of the parts like bonnet, panel, or engine would be a project. The integration of all the parts (projects) would make a complete automobile (program).

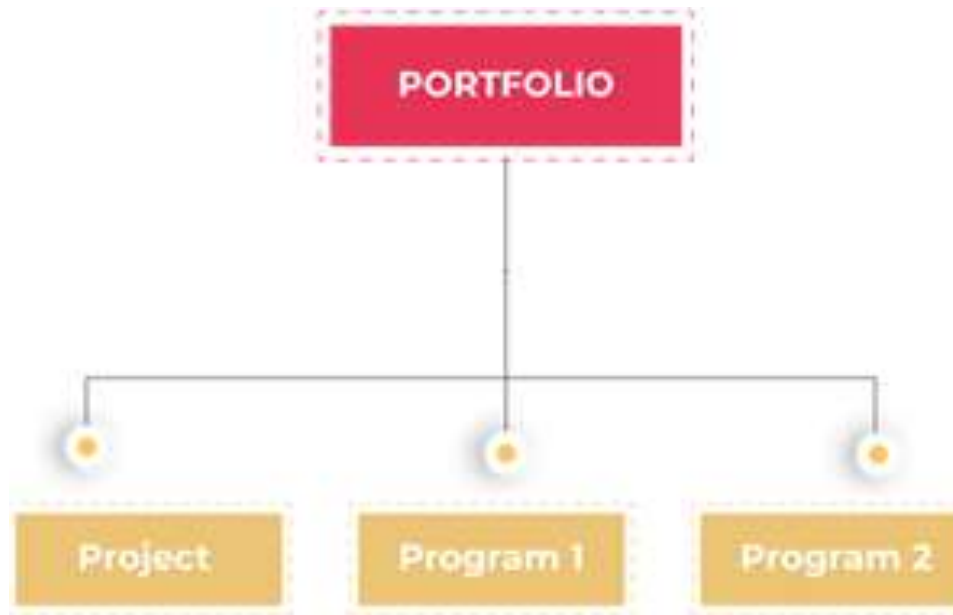


Project, Program and Portfolio Management

Portfolio

A portfolio refers to a collection of projects, programs and operations managed as a group to achieve strategic objectives.

For example, take the case of an automobile company. The unique designs of different cars will represent programs, and the design of cars, in general, will represent portfolio.



Project Selection Methods

There are various project selection methods practiced by the modern business organizations.

There are two categories of project selection methods:

1. Benefit Measurement Methods
2. Constrained Optimization Methods

These methods have different features and characteristics. Therefore, each selection method is best for different organizations.

1- Benefit Measurement Methods

Benefit Measurement is a project selection technique based on the present value of estimated cash outflow and inflow. Cost benefits are calculated and then compared to other projects to make a decision.

The techniques that are used in Benefit Measurement are as follows:

- Cost Benefit Ratio (CBR)
- Return on Investment (ROI)
- Payback Period (PP)
- Discounted Cash flow (PV, FV)
- Net Present Value (NPV)
- Internal Rate of Return (IRR)

2- Constrained Optimization Methods

Constrained Optimization Methods, also known as the Mathematical Model of Project Selection, are used for larger projects that require complex and comprehensive mathematical calculations.

The techniques that are used in Constrained Optimization Methods are as follows:

- Linear Programming
- Non-Linear Programming
- Integer Programming
- Dynamic Programming
- Multiple objective Programming

Cost-Benefit Ratio

CBA is a process or tool to support decision making in projects.

How to Calculate Cost-Benefit Ratio (CBR)

$\text{Cost-Benefit Ratio} = (\text{Total Earning} / \text{Total Cost})$

Example:

An example of Cost-Benefit Analysis includes Cost-Benefit Ratio where suppose there are two projects as follows:

Project # 1	Project # 2
Total cost of \$8,000	Costs of \$11,000
Earning total benefits of \$ 12,000	Earning benefits of \$ 20,000
Cost-Benefit Ratio = $(\$12,000/\$8,000)$	Cost-Benefit Ratio = $(\$20,000/\$11,000)$
Cost-Benefit Ratio = 1.5	Cost-Benefit Ratio = 1.81

By applying cost-benefit analysis of the first project is **1.5** and the ratio of the second project is **1.81** which means project two is feasible being having high cost-benefit ratio.

Return on investment (ROI)

Return on investment (ROI) refers to the percentage of increase or decrease of an investment within a year.

$$\text{How to Calculate Return on Investment (ROI)} == \text{ROI} = \frac{\text{Net Return on Investment}}{\text{Cost of Investment}} \times 100\%$$

Example:

You are a house flipper. You purchased a house at the courthouse auction for \$75,000 and spent \$35,000 in renovations. After sales, expenses, and commission, you netted \$160,000 on the sale of the renovated house. What is the ROI?

Solution:

Your net profit is going to be what you netted (\$160,000) minus what you spent (\$75,000 + \$35,000), so it is **\$50,000**. Your total investment is also what you spent (\$75,000 + \$35,000), which is **\$110,000**.

$$\text{ROI} = (\text{Net Return on Investment} / \text{Total Cost of Investment}) * 100$$

$$\text{ROI} = (50,000 / 110,000) * 100$$

$$\text{ROI} = .45 * 100$$

$$\text{ROI} = 45\%$$

Payback Period

The payback period is the length of time it takes the company to recoup the initial costs of producing the product, service, or result of the project. This method compares the initial investment to the cash inflows expected over the life of the product, service, or result.

Example

For example, say the initial investment on a project is \$200,000, with expected cash inflows of \$25,000 per quarter every quarter for the first two years and \$50,000 per quarter from then on.

The payback period is two years and can be calculated as follows:

Initial investment (Cash outflow) = \$200,000

Cash inflows = $\$25,000 \times 4$ (quarters in a year) = \$100,000 per year total inflow

Initial investment (\$200,000) – year 1 inflows (\$100,000) = \$100,000 remaining balance

Year 1 inflows remaining balance – year 2 inflows = \$0

Total cash flow year 1 and year 2 = \$200,000

The payback is reached in two years.

Discounted Cash Flows

As I just stated, money received in the **future** is worth less than money received **today**. The reason for that is the time value of money.

Example

If I borrowed **Rs.2, 000** from you today and promised to pay it back in **three years**, you would expect me to pay interest in addition to the original amount borrowed. Suppose that interest rate at 5% per year. Here's the formula for future value calculations: $FV = PV(1 + i)^n$

PV is Present Value,

FV is Future Value

i is the interest rate (as a decimal, so 0.10, not 10%)

n is the number of years

$$FV = \text{Rs. } 2,000(1 + .05)^3$$

$$FV = \text{Rs. } 2,000(1.157625)$$

$$FV = \text{Rs. } 2,315.25$$

Discounted Cash Flows

The discounted cash flow technique compares the value of the future cash flows of the project to today's Rs. To calculate discounted cash flows, you need to know the value of the investment in today's terms, or the PV. PV is calculated as follows:

$$PV = FV / (1 + i)^n$$

This is the reverse of the FV formula talked about earlier. So, if you ask the question, “What is Rs.2,315.25 in three years from now worth today given a 5 percent interest rate?” you'd use the preceding formula. Let's try it:

$$PV = \text{Rs. } 2,315.25 / (1 + .05)^3$$

$$PV = \text{Rs. } 2,315.25 / 1.157625$$

$$PV = \text{Rs. } 2,000$$

Rs. 2,315.25 in three years from now is worth \$2,000 today

Net Present Value (NPV)

Net present value (NPV) is the difference between the present value of cash inflows and the present value of cash outflows over a period of time. NPV is used in capital budgeting and investment planning to analyze the profitability of a projected investment or project.

When to Accept / Reject Project:

- ✓ IF $NPV > 0$, Project Accept
 - ✓ IF $NPV < 0$, Project Reject
-
- If NPV has the bigger the number, the more potential the project has.
 - If you end up with a negative number, your project won't be profitable.

Net Present Value (NPV)

Example:

The following table shows you what this might look like for a project with initial investment **\$97000**. That will be released over three years with predicted future values for each year with Interest rate is 6%.

Table Finding the Net Present Value		
Year	Future Value	Present Value
1	\$35,000	\$33,019
2	\$48,000	\$42,720
3	\$81,000	\$68,009
Sum PV:		\$143,748

NPV (Sum of PV – Investment) \$46,748

NPV > 0, This project will be accepted.

Internal Rate of Return (IRR)

Internal Rate of Return is the interest rate that makes the Net Present Value zero. And that "guess and check" method is the common way to find it.

Example:

Invest \$2,000 now, receive 3 yearly payments of \$100 each, plus \$2,500 in the 3rd year.

Let us try 10% interest rate:

Now: $PV = -\$2,000$

Year 1: $PV = \$100 / 1.10 = \90.91

Year 2: $PV = \$100 / 1.10^2 = \82.64

Year 3: $PV = \$100 / 1.10^3 = \75.13

Year 3 (final payment): $PV = \$2,500 / 1.10^3 = \$1,878.29$

$NPV = (\text{Sum of PV} - \text{Investment})$

NPV = $\$90.91 + \$82.64 + \$75.13 + \$1,878.29 - \$2,000 = \126.97

NPV at 10% discount rate = \$126.97

Since NPV is greater than zero we have to increase discount rate.

Internal Rate of Return (IRR)

Let's try a better guess, say **12% interest rate**:

Now: $PV = -\$2,000$

Year 1: $PV = \$100 / 1.12 = \89.29

Year 2: $PV = \$100 / 1.122 = \79.72

Year 3: $PV = \$100 / 1.123 = \71.18

Year 3 (final payment): $PV = \$2,500 / 1.123 = \$1,779.45$

$NPV = (\text{Sum of PV} - \text{Investment})$

$NPV = \$89.29 + \$79.72 + \$71.18 + \$1,779.45 - \$2,000 = \19.64

$NPV \text{ at } 12\% \text{ discount rate} = \19.64

But it is still greater than zero we have to further increase the discount rate,

so close. Maybe 12.4% ?

Internal Rate of Return (IRR)

Let us try 12.4% interest rate:

Now: $PV = -\$2,000$

Year 1: $PV = \$100 / 1.124 = \88.97

Year 2: $PV = \$100 / 1.1242 = \79.15

Year 3: $PV = \$100 / 1.1243 = \70.42

Year 3 (final payment): $PV = \$2,500 / 1.1243 = \$1,760.52$

$NPV = (\text{Sum of PV} - \text{Investment})$

NPV = $\$88.97 + \$79.15 + \$70.42 + \$1,760.52 - \$2,000 = -\0.94

Since NPV is fairly close to zero at 12.4% value of interest rate, therefore

Let us stop there and say the Internal Rate of Return is **12.4%**